

MTH 419 Introduction to Abstract Algebra Syllabus - Fall 2026

Lectures: Tue, Thu 11:00-12:20 AM, Math Building 150

Recitations: Fri 1:00-1:50 PM, Baldy 200G

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Course Resources:

- **Course Website:** mth419.ubmath.info. See this website for an up-to-date version of this syllabus, course lecture notes, homework assignments etc.
- **Lecture notes.** Skeletal lecture notes for each week will be posted on the course website. I will annotate these notes during lectures. Please print the notes and bring them to class to follow along.
- **Piazza.** We be using Piazza for course-related communications. The link to the Piazza webpage for this course is
<https://piazza.com/buffalo/fall2026/mth419/home>
If you have a question or comment related to the course please post it on Piazza. This will help other students who may deal the same issue. If you know the answer to a question somebody else posted on Piazza, please answer it. If you have a personal question (concerning your grade etc.), you can either send to me a private message on Piazza or contact me by e-mail.
- **Textbook.** Joseph A. Gallian, *Contemporary Abstract Algebra*.
This book is currently in the 11th edition, but any older edition is just as fine - they all cover the same material and changes between various editions are mostly cosmetic.
- **UBLearns.** UBLearns will be used to collect homework assignments.

Homework. Homework problems will be assigned every week. The goal of homework is to let you practice with the course material and to provide you with feedback on your work. With this mind, you will get back homework assignments with comments on your solutions, but each honest attempt at a solution will receive full credit - regardless if it is fully correct or not. The only rule is that you must fully understand solutions you are submitting. Occasionally, I may request to meet with you so you can explain your work in person. If you will not be able to do it, you will receive no credit for the homework assignment, and additionally your final course grade will be lowered by one letter grade (from A to A-, from A- to B+ etc.) for each instance.

Exams:

Midterm 1	Thursday, October 8	11:00-12:20 PM	Math Building 150
Midterm 2	Thursday, November 12	11:00-12:20 AM	Math Building 150
Final Exam	Thursday, December 10	11:45 AM-2:45 PM	Math Building 150

Grading. You will receive a letter grade for each exam. Homework will be assigned letter grades on the following scale: 100-88% of the maximum possible score = A, 87-75% = B, 74-62% = C, 61-49% = D, 49-0% = F. The final course grade will be computed as a weighted average of all these latter grades, with weights given as follows:

Homework	20%
Midterm 1	23%
Midterm 2	23%
Final Exam	34%

Learning outcomes. The goal of this course is to introduce basic notions and methods of abstract algebra. After completing it, you should have working knowledge of the following topics:

- Definition and elementary properties of groups.
- Cyclic groups, dihedral groups, symmetric groups.
- Homomorphisms and isomorphisms of groups.
- Subgroups
- Direct products of groups.
- Cosets, Lagrange theorem.
- Group actions on sets.
- Normal subgroups and quotient groups.
- Isomorphism theorems.
- Classification of finitely generated abelian groups.
- Rings.
- Ideals of rings and quotient rings.
- Ring homomorphisms.
- Integral domains.
- Polynomial rings.
- Fields.

Prerequisites. The main expectation is that students taking this course are familiar with reading and writing mathematical proofs to the extent that is covered in MTH 311. We will also use some notions related to sets and functions (also typically covered in MTH 311), as well as some elementary number theory (divisibility of integers, the greatest common divisor etc.). From time to time we will also use linear algebra (matrices, determinants, vectors etc.). These topics are covered in MTH 309.

Math Help Center. The Math Help Center is an additional place (beside lectures, recitations, and office hours) where you can seek help with questions related to this course. The Math Help Center is located in room 110 of the Mathematics Building and it is staffed by math graduate students. You don't need an

appointment to visit it, you can stop by at any time during its opening hours. The current schedule of the Math Help Center is posted here: www.buffalo.edu/cas/math/ug/help-center.html.

Incomplete grades. See the UB Catalog for the UB incomplete grades policy:

catalogs.buffalo.edu/content.php?catoid=11&navoid=571#incomplete-grades

Academic integrity. See the UB Catalog for the UB academic integrity policy:

catalogs.buffalo.edu/content.php?catoid=11&navoid=571#academic-integrity

Accessibility resources. If you have any disability that requires reasonable accommodations to enable you to participate in this course, please contact the Office of Accessibility Resources in 60 Capen Hall at 716-645-2608 and the instructor of this course during the first week of class. The office will provide you with information and review appropriate arrangements for reasonable accommodations, as outlined on the Accessibility Resources website:

www.buffalo.edu/studentlife/who-we-are/departments/accessibility.html